MONASH UNIVERSITY • MCS16

GNN-Based Multi-Omics Integration for Predicting Cancer Drug Response

MCS16

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WASIF HUSSAIN

Agenda

01 Overview & Updated SMART Goals

Wasif Hussain



02 Methodology, Roles & Progress Tracking

Aditya Ansh



03 Current Issues & Risk Management

Charmaine Chee



04 MVP Demo & What's Next

Shaahid Marikar



WASIF HUSSAIN

Problem Recap & Updated SMART Goals

WHY THIS MATTERS

Cancer heterogeneity defeats one-size-fits-all drugs.

GNNs preserve biological network topology that flat feature vectors discard.

Benchmark: MoGraphDRP RMSE 0.6622 (random split)

SMART GOAL 01

Close the architecture gap by Week 9

0.6622	**MoGraph RMSE**	**0.8971**	**GNN (random split)**
1.2843	**flat MLP**	**1.3301**	**tuned GNN**

SMART GOAL 02

Deploy a functional multi-omics prototype

CSV input → IC50 prediction → feature importance
Deliverables: integrated GNN engine + standalone UI dashboard

Toolchain

ADITYA ANSH

Notion
Project Governance

MANAGEMENT

Single Source of Truth

- Sprint planning boards & user-story backlogs
- Requirements Traceability Matrix (RTM)
- Risk register & centralized meeting notes

GitHub
Source & Quality Control

CODE REVIEW

Branch & PR Workflow

- Branch-per-member isolated feature development
- Mandatory pull-request code reviews
- Experiment results logged in commit history & docs

Google Colab
Model Development

COMPUTE

Cloud GPU Execution

- GPU-backed model training & pipeline execution
- Hyperparameter tuning & ablation runs
- Workloads managed within local VRAM bounds

Communication Stack
Real-time Sync & Official Channels

COMMUNICATION

WhatsApp
- Daily quick team check-ins
- Real-time blocker flagging & resolution

Gmail
- Formal supervisor correspondence
- Milestone reporting & external reviews

Operations & Assets
Time Allocation & File Repository

OPERATIONS

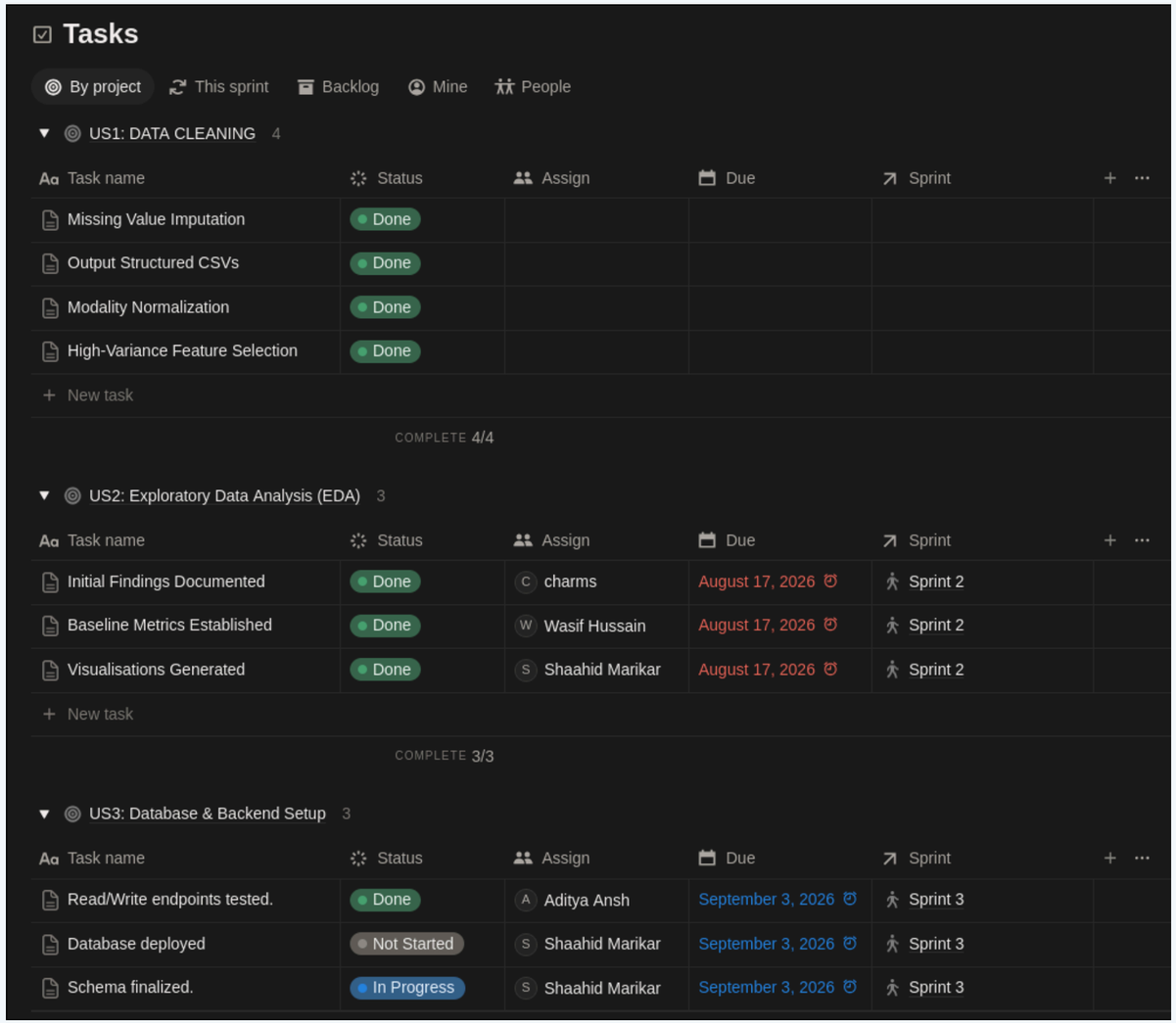
Google Calendar
- Deadline Management

Google Drive
- Shared docs & large assets
- Non-code deliverable storage

ADITYA ANSH

Progress Tracking - Notion Hub

- User Story board
- Gives a view of overall sprints across all user stories
- 3 user stories done, 1 in planning



Tasks

By project | This sprint | Backlog | Mine | People

US1: DATA CLEANING 4

Task name	Status	Assign	Due	Sprint
Missing Value Imputation	Done			
Output Structured CSVs	Done			
Modality Normalization	Done			
High-Variance Feature Selection	Done			

New task

COMPLETE 4/4

US2: Exploratory Data Analysis (EDA) 3

Task name	Status	Assign	Due	Sprint
Initial Findings Documented	Done	C charms	August 17, 2026	Sprint 2
Baseline Metrics Established	Done	W Wasif Hussain	August 17, 2026	Sprint 2
Visualisations Generated	Done	S Shaahid Marikar	August 17, 2026	Sprint 2

New task

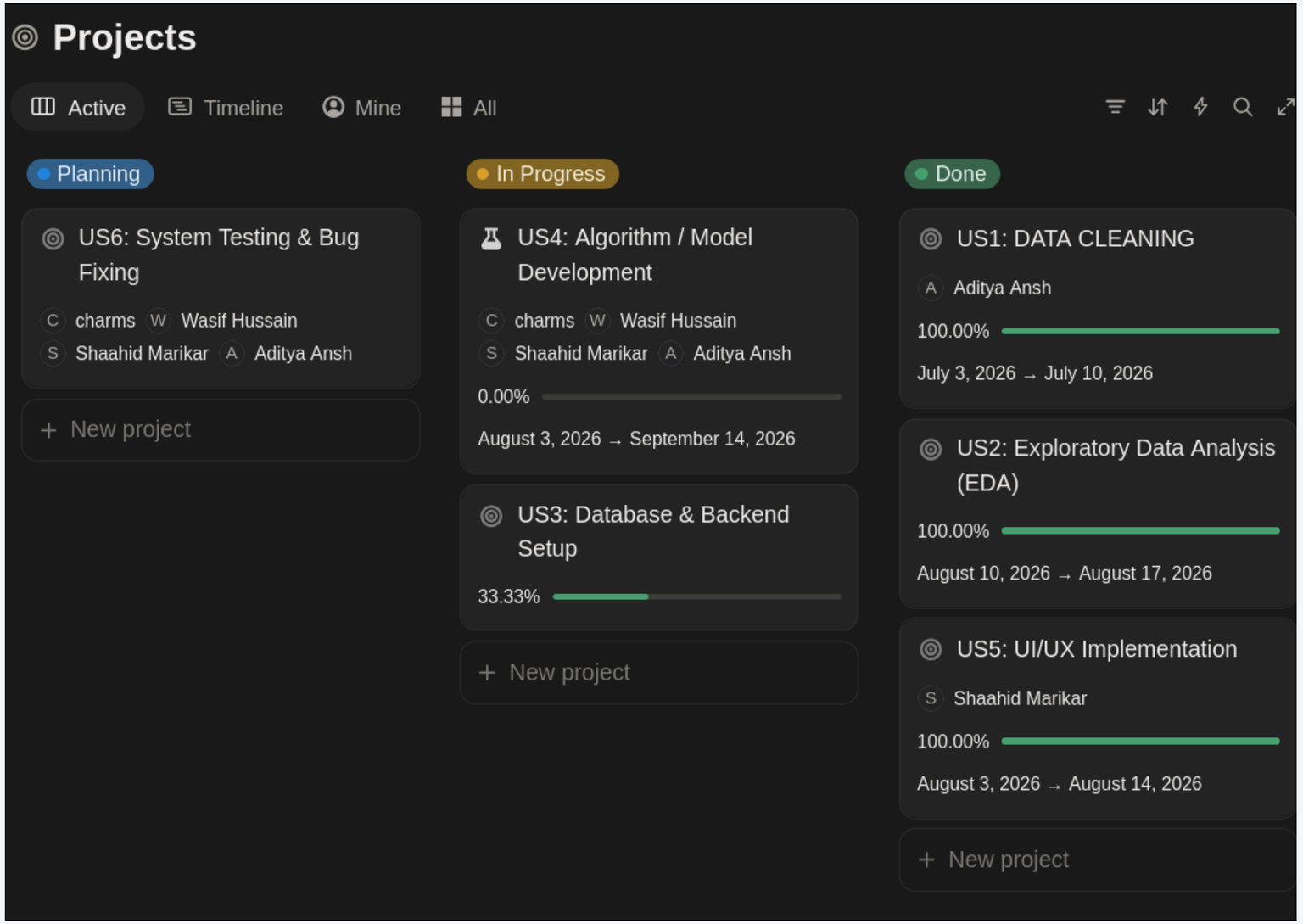
COMPLETE 3/3

US3: Database & Backend Setup 3

Task name	Status	Assign	Due	Sprint
Read/Write endpoints tested.	Done	A Aditya Ansh	September 3, 2026	Sprint 3
Database deployed	Not Started	S Shaahid Marikar	September 3, 2026	Sprint 3
Schema finalized.	In Progress	S Shaahid Marikar	September 3, 2026	Sprint 3

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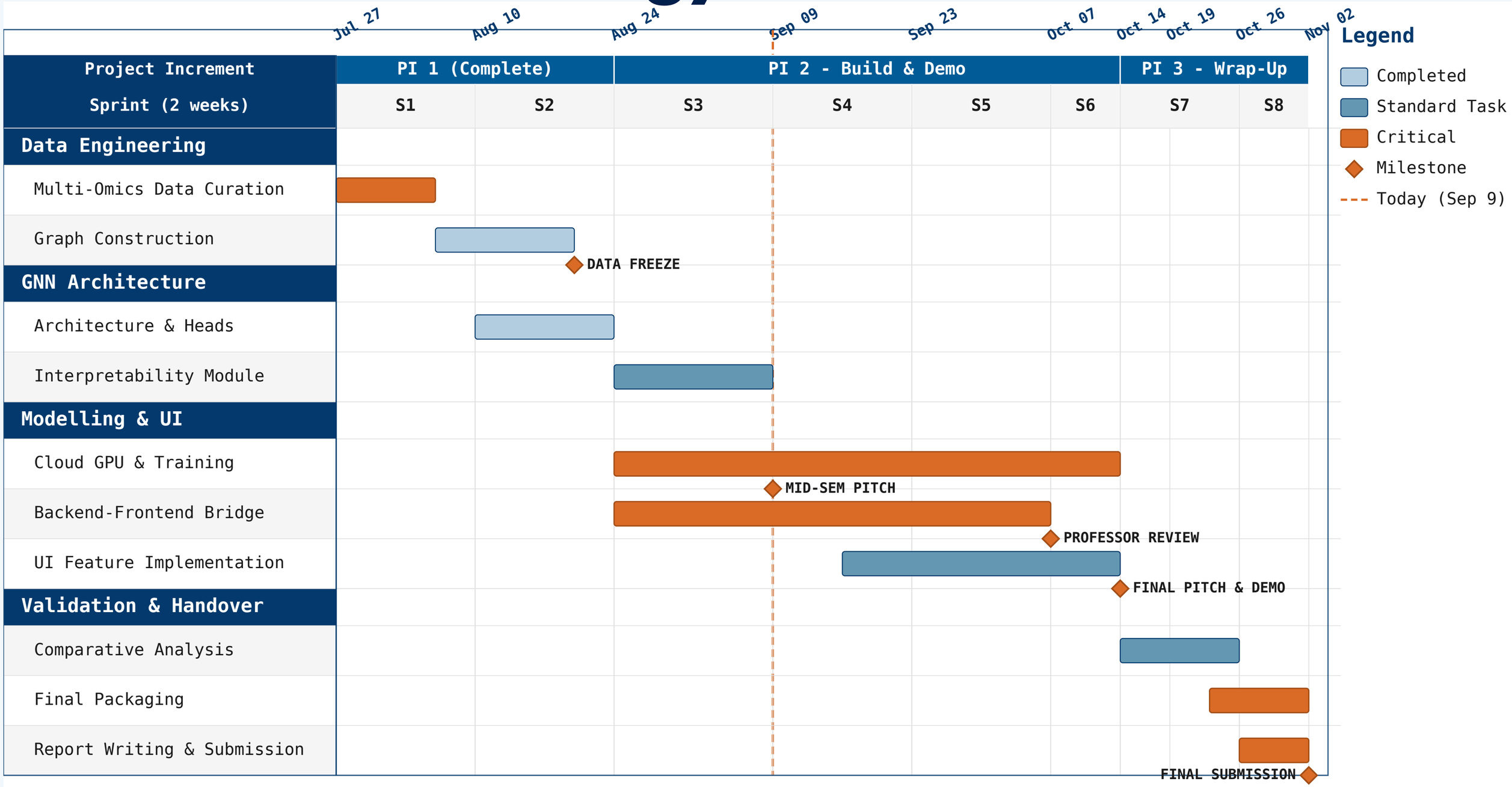
Progress Tracking - Notion Hub



- Task-level backlog broken down by User Story
- Tasks carry:
  - Owner
  - Due Date
  - Sprint Tag
  - Status

Methodology & Gantt Chart

ADITYA ANSH



Progress Tracking - RTM

ADITYA ANSH

Req	Description	Categories	Type	Source
R02	Delete user uploaded files after a prediction session ends, ensuring no sensitive research data is retained.	Security, Privacy	NFR	Developer
R04	Display a molecular profile summary of the top 5 mutations	UI, Visualisation	FR	Oncologist

5 / 7

requirements complete

1 / 3

major milestones achieved

CHARMAINE CHEE

Top-3 Issues Currently Being Tackled

01

Benchmark comparability

0.6622 random vs 1.3301 grouped; matched protocol: 0.8971.
60% protocol / 40% architecture.
Always report split + RMSE.

02

Confidence calibration

Confidence scores use a fixed formula based on dropout uncertainty and validation RMSE, without calibration to actual prediction errors. This may result in similar confidence scores for predictions with very different levels of reliability.

03

Feature dimensionality overload

Tri-omics integration yields >40,000 features for only 532 viable cell lines, creating a high risk of overfitting. Currently evaluating PCA and autoencoder bottlenecks to compress the feature space before training.

CHARMAINE CHEE

Risk Register

NEW RISK	**UI blocking & high latency during model inference**	Implement asynchronous processing
MONITORED	**Fusion may not beat simple baselines**	Cross-attention + continued GNN tuning
ANTICIPATED	**Close GNN-to-MLP performance gap**	Improve graph construction

LIVE DEMO

MVP DEMO

Desktop dashboard walkthrough

01 Dataset Upload

02 Model Execution Log

03 Final Results

Real prediction from the best HeteroIC50GNN checkpoint. One function highlighted; full pipeline narration deferred.

SHAAHID MARIKAR



SHAAHID MARIKAR

Next Steps & Priorities

Improve Model Performance & Usability

01 **Implement Atomic Level Graph**

Add layer depth to improve prediction

02 **Improve User Understanding**

Add tooltip, model visualisations & notifications

CONTINUE

GNN + protein–protein interaction track

Sprint priorities and KIV items remain visible on the team Notion board.

Thank You!

References

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